ELEC50011A - MATHEMATICS 2A-2025-SOLUTIONS

1. a) Let a mapping from the $z-$plane to the $w-$ plane be given by

$$w(z) = \frac{z-1}{z+i}.$$

Determine the image in the $w-$plane of the $x-$axis in the $z-$plane. Which curve in the $z-$plane maps to the line $u = v$ in the $w-$plane? . [5]

SOLUTION Begin with $w = \dfrac{x-1+iy}{x+i(y+1)}$ and multiply appropriately to

make the denominator real:

$$w = \frac{(x-1+iy)[x-i(y+1)]}{[x+i(y+1)][x-i(y+1)]} = \frac{(x-1)x+y(y+1)+i[xy-(x-1)(y+1)]}{x^2+(y+1)^2} = u+iv$$

where

$$u = \frac{x^2-x+y^2+y}{x^2+(y+1)^2} \quad \text{and} \quad v = \frac{y-x+1}{x^2+(y+1)^2}$$

Setting $u = v \Rightarrow u - v = 0$ we have

$$\frac{x^2-x+y^2+y}{x^2+(y+1)^2} - \frac{y-x+1}{x^2+(y+1)^2} = \frac{x^2-x+y^2+y-(y-x+1)}{x^2+(y+1)^2} = \frac{x^2+y^2-1}{x^2+(y+1)^2} = 0.$$

Hence $x^2+y^2 = 1$, the unit circle in the $z-$plane, maps to the line $u = v$ in the $w-$plane.

On the $x-$axis in the $z-$plane, we have $y = 0$, giving

$$u = \frac{x^2-x}{x^2+1} \quad \text{and} \quad v = \frac{-x+1}{x^2+1}$$

Spot that the numerator of both u, v hve the factor $x-1$ and add

$$u^2+v^2 = \frac{x^2(x-1)^2+(x-1)^2}{(x^2+1)^2} = \frac{(1-x)^2}{x^2+1} = \frac{x^2-2x+1}{x^2+1} = u+v$$

so we have the circle

$$u^2-u+\frac{1}{4}+v^2-v+\frac{1}{4} = \left(u-\frac{1}{2}\right)^2+\left(v-\frac{1}{2}\right)^2 = \frac{1}{2}$$

The $x-$axis in the $z-$plane maps to a circle with centre $(1/2, 1/2)$ and radius $1/\sqrt{2}$, in the $w-$plane.

b) Show that the real integral

$$\int_0^{2\pi} e^{\cos\theta}\,\cos(\sin\theta)\,d\theta \qquad (1)$$

can be written in terms of the imaginary part of the complex contour integral

$$\oint_C \frac{e^z}{z}\,dz \qquad (2)$$

where C is the unit circle in the complex plane. Use Cauchy's residue theorem to evaluate (2), and hence obtain (1). [6]

SOLUTION

On the unit circle, $z = e^{i\theta} = \cos\theta + i\sin\theta$, where θ varies from 0 to 2π. We use

$$\frac{dz}{d\theta} = ie^{i\theta} = iz \Rightarrow dz = izd\theta$$

and substitute into (2), which gives

$$\oint_C \frac{e^z}{z}\,dz = \int_0^{2\pi} \frac{e^{\cos\theta+i\sin\theta}}{z} izd\theta = i\int_0^{2\pi} e^{\cos\theta}e^{i\sin\theta}d\theta = i\int_0^{2\pi} e^{\cos\theta}[\cos(\sin\theta)+i\sin(\sin\theta)]\,d\theta$$

$$= i\int_0^{2\pi} e^{\cos\theta}\cos(\sin\theta)d\theta - \int_0^{2\pi} e^{\cos\theta}\sin(\sin\theta)\,d\theta$$

and we see that the integral constituting the imaginary part, matches (1), as required.

Now use Cauchy's Residue Theorem on (2). The function $f(z) = e^z/z$ has a simple pole at $z = 0$, inside C, with residue

$$\lim_{z\to 0}(z-0)f(z) = \lim_{z\to 0} e^z = 1.$$

By Cauchy's residue theorem, the integral is equal to
$2\pi i \times$ (Sum of residues inside C) $= 2\pi i$.
The imaginary part of this is 2π, and hence the value of (1).

c) To obtain the value of the real, improper integral

$$I = \int_{-\infty}^{+\infty} \frac{\cos x}{(x^2+4)^2}\, dx\,,$$

we proceed as follows:

(i) The function $F(z) = \dfrac{e^{iz}}{(z^2+4)^2}$ has two double poles at $z = \pm 2i$. Obtain the residue of $F(z)$ at the pole in the upper half-plane. [3]

(ii) Choose an appropriate semi-circular contour C lying on the real axis and in the upper half-plane to evaluate the complex integral

$$\oint_C F(z)\, dz\,.$$

Hence, or otherwise, find the value of I. Be sure to explain your reasoning! [6]

SOLUTION

(i) We find the residues for the double pole of $F(z)$ at $z = 2i$:

$$\begin{aligned}\text{Res of } F \\ @z = 2i\end{aligned} : \lim_{z\to 2i} \frac{d}{dz}\left[(z-2i)^2 \frac{e^{iz}}{(z^2+4)^2}\right] = \lim_{z\to 2i} \frac{d}{dz}\left[\frac{e^{iz}}{(z+2i)^2}\right]$$

$$= \lim_{z\to 2i} \frac{ie^{iz}(z+2i)^2 - e^{iz}2(z+2i)}{(z+2i)^4} = \frac{ie^{i(2i)}(4i)^2 - e^{i(2i)}2(4i)}{(4i)^4} = -\frac{3}{2}e^{-2}i\,.$$

(ii) The semi-circular contour C (sketch below) includes the double pole at $z = 2i$, so using Cauchy's Residue Theorem, we obtain

$$\oint_C F(z)dz = 2\pi i[\text{Sum of residues inside } C] = 2\pi i\left(-\frac{3}{2}e^{-2}i\right) = 3\pi e^{-2}\,.$$

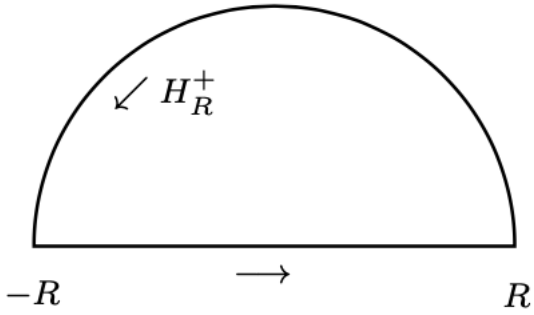
(iii) We divide the semi-circular contour into two segments: on the axis, over the real interval $[-R, R]$ and the semi-circular arc H_R in the upper half-plane, see sketch. Hence,

$$\oint_C F(z)dz = \int_{-R}^{R} \frac{e^{ix}}{(x^2+4)^2}\, dx + \int_{H_R} \frac{e^{iz}}{(z^2+4)^2}\, dz = 3\pi e^{-2}$$

In the limit $R \to \infty$, we apply Jordan's Lemma to the second integral which vanishes, given that the only singularities are poles, and with $m > 0$, the denominator behaves as $R^{-4} \to 0$. This leaves

$$\int_{-\infty}^{+\infty} \frac{e^{ix}}{(x^2+4)^2}\, dx = \int_{-\infty}^{+\infty} \frac{\cos x}{(x^2+4)^2}\, dx + i\int_{-\infty}^{+\infty} \frac{\sin x}{(x^2+4)^2}\, dx = 3\pi e^{-2}$$

The second integral is equal to zero, as the integral of an odd function over a symmetric interval. Hence $I = 3\pi e^{-2}$.


H_R^+
$-R$
R

2. a) A harmonic function is given as $v(x,y) = \cos y \cosh x + 2\sin y \sinh x$. Find its harmonic conjugate $u(x,y)$, and hence the analytic function $f(z)$ where $f(z) = u + iv$. [5]

SOLUTION We integrate the Cauchy-Riemann equations:

$$u_x = v_y = -\sin y \cosh x + 2\cos y \sinh x \Rightarrow$$

$$u = \int v_y \, dx = \int -\sin y \cosh x + 2\cos y \sinh x \, dx = -\sin y \sinh x + 2\cos y \cosh x + A(y)$$

and

$$u_y = -v_x = -\cos y \sinh x - 2\sin y \cosh x \Rightarrow$$

$$u = -\int v_x \, dy = -\int \cos y \sinh x + 2\sin y \cosh x \, dy = -\sin y \sinh x + 2\cos y \cosh x + B(x)$$

where A, B are arbitrary functions. Equating both, we have $A = B =$ *constant*, and $u = -\sin y \sinh x + 2\cos y \cosh x + C$.

Hence $f(z) = u + iv =$
$-\sin y \sinh x + 2\cos y \cosh x + i\cos y \cosh x + 2i \sin y \sinh x + C$
At this point you need to spot a pattern:
$= 2(\cos y \cosh x + i \sin y \sinh x) + i(\cos y \cosh x + i \sin y \sinh x) + C$
$= 2[\cos y \cos(ix) + \sin y \sin(ix)] + i[\cos y \cos(ix) + \sin y \sin(ix)] + C$
$= (2+i)\cos(y - ix) = (2+i)\cos(iz) + C.$
Hence $f(z) = (2+i)\cos(iz) + C$

b) An invertible matrix is given as

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Use Gram-Schmidt orthogonalization and the columns of A to find an orthonormal basis for $\mathbb{R}^3$. Use the basis to find an orthogonal matrix Q and an upper-triangular matrix R, so that $A = QR$: the $QR-$factorization of A. [6]

Solution: Apply Gram-Schmidt. Let $\underline{\mathbf{a}}_i$ be column i of A. Then

$$\underline{\mathbf{x}}_1 \;=\; \underline{\mathbf{a}}_1 \;=\; \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \underline{\mathbf{x}}_2 = \underline{\mathbf{a}}_2 - \frac{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{a}}_2}{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{x}}_1}\underline{\mathbf{x}}_1 \;=\; \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

and we take $\underline{\mathbf{x}}_2 = (1,-1,2)^T$ to simplify the next calculation:

$$\underline{\mathbf{x}}_3 = \underline{\mathbf{a}}_3 - \frac{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{a}}_3}{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{x}}_1}\underline{\mathbf{x}}_1 - \frac{\underline{\mathbf{x}}_2 \cdot \underline{\mathbf{a}}_3}{\underline{\mathbf{x}}_2 \cdot \underline{\mathbf{x}}_2}\underline{\mathbf{x}}_2$$

$$= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}}{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \frac{\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}}{\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{6}\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \;=\; \frac{2}{3}\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

Now normalize to get the orthonormal basis $\underline{\mathbf{q}}_i = \frac{1}{|\underline{\mathbf{x}}_i|}\underline{\mathbf{x}}_i$

$$\underline{\mathbf{q}}_1 = \frac{1}{|\underline{\mathbf{x}}_1|}\underline{\mathbf{x}}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \underline{\mathbf{q}}_2 = \frac{1}{|\underline{\mathbf{x}}_2|}\underline{\mathbf{x}}_2 = \frac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \underline{\mathbf{q}}_3 = \frac{1}{|\underline{\mathbf{x}}_3|}\underline{\mathbf{x}}_3 = \frac{1}{\sqrt{3}}\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

Now find the orthogonal matrix Q: the orthonormal vectors give the columns of Q, so that

$$Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

As Q is orthogonal we have $Q^{-1} = Q^T$ so

$$A = QR \Rightarrow R = Q^T A = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}^T \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{pmatrix}.$$

c) Find an orthogonal matrix S and a diagonal matrix Λ, so that $A = S\Lambda S^T$: the orthogonal diagonalization of the 3×3matrix

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Solution: For the orthogonal diagonalization, we first need eigenvalues and eigenvectors of A. Find eigenvalues as $\det(A - \lambda I) = 0$, so

$$\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} \underset{\substack{first\ R_2 - R_3 \\ then\ C_2 + C_3}}{=} \begin{vmatrix} -\lambda & 2 & 1 \\ 0 & 0 & \lambda + 1 \\ 1 & 1 - \lambda & -\lambda \end{vmatrix} = 0$$

as determinant does not change under row/col operations. Now expand by the second row and get

$\det(A - \lambda I) = (\lambda + 1)[-\lambda(1 - \lambda) - 2] = (\lambda + 1)^2(\lambda - 2) = 0$. We have a repeated eigenvalue, so will need to find two orthogonal eigenvectors for $\lambda = -1$:

$$(A + I)\underline{\mathbf{x}} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow x + y + z = 0$$

letting $y = 0, 1$ and $z = 1, 0$, gives independent eigenvectors:

$$\underline{\mathbf{a}}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \underline{\mathbf{a}}_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

We need orthogonal eigenvectors, so we use Gram-Schmidt, one step

$$\underline{\mathbf{x}}_1 = \underline{\mathbf{a}}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \text{ and } \underline{\mathbf{x}}_2 = \underline{\mathbf{a}}_2 - \frac{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{a}}_2}{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{x}}_1} \underline{\mathbf{x}}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/2 \\ -1/2 \\ 1 \end{pmatrix}$$

The single eigenvalue $\lambda = 2$ gives

$$(A - 2I)\underline{\mathbf{x}} = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and with two row operations we have $x - 2y + z = 0$ and $-y + z = 0$, giving eigenvector

$$\underline{\mathbf{x}}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

The normalized eigenvectors form the diagonalizing matrix

$$S = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \text{ with diagonal matrix } \Lambda = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

so that $A = S\Lambda S^T$ is the orthogonal diagonalization.

d) Let λ, γ be distinct eigenvalues of the symmetric matrix A, with respective eigenvectors $\underline{\mathbf{u}}, \underline{\mathbf{v}}$. Prove that $\underline{\mathbf{u}}, \underline{\mathbf{v}}$ are orthogonal vectors. [4]

Solution:

If λ, γ are eigenvalues of A, eigenvectors $\underline{\mathbf{u}}, \underline{\mathbf{v}}$, we can write

$$\lambda \underline{\mathbf{u}} = A\underline{\mathbf{u}} \quad \text{and} \quad \gamma \underline{\mathbf{v}} = A\underline{\mathbf{v}}$$

Hence consider:

$(A\underline{\mathbf{v}})^T \underline{\mathbf{u}} = (\gamma \underline{\mathbf{v}})^T \underline{\mathbf{u}} = \gamma(\underline{\mathbf{v}}^T \underline{\mathbf{u}})$.

Now write the above lhs as

$(A\underline{\mathbf{v}})^T \underline{\mathbf{u}} \;=\; (\underline{\mathbf{v}}^T A^T)\underline{\mathbf{u}} \;=\; \underline{\mathbf{v}}^T (A\underline{\mathbf{u}})$, given that A is symmetric,

$= \; \underline{\mathbf{v}}^T (\lambda \underline{\mathbf{u}}) \;=\; \lambda(\underline{\mathbf{v}}^T \underline{\mathbf{u}})$.

Equating the two expressions we have

$$\lambda(\underline{\mathbf{v}}^T \underline{\mathbf{u}}) \;=\; \gamma(\underline{\mathbf{v}}^T \underline{\mathbf{u}}) \;\Rightarrow (\lambda - \gamma)\underline{\mathbf{v}}^T \underline{\mathbf{u}} \;= 0 \, .$$

Given that the eigenvalues are distinct, $\lambda - \gamma \neq 0$, leaving $\underline{\mathbf{v}}^T \underline{\mathbf{u}} \;= 0$, so they are orthogonal, as required.

e) Given

$$A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \\ 2 & 2 \end{pmatrix} ,$$

its Singular Value Decomposition (SVD) is $A = U\Sigma V^T$.

Explain *in words* how the first stage of finding the SVD of A involves the orthogonal diagonalization of a symmetric matrix, and find the SVD of A. [10]

Solution:

As A is 3×2 we require the smaller symmetric matrix $A^T A$ which will be 2×2. In the SVD, Σ has all off-diagonal elements zero, and U, V are orthogonal matrices. Substituting

$$A^T A = (U\Sigma V^T)^T (U\Sigma V^T) = V\Sigma^T U^T U \Sigma V^T$$

and as U is an orthogonal matrix, $U^T U$ is the identity matrix, leaving $A^T A = V\Sigma^T \Sigma V^T$. The 3×2 matrix Σ has the singular values σ_i on the diagonal, so the 2×2 matrix $\Lambda = \Sigma^T \Sigma$ has diagonal elements σ_i^2. As V is orthogonal, we have $A^T A = V \Lambda V^T$, the orthogonal diagonalization of $A^T A$.

Now to find the SVD, we begin with

$$A^T A = \begin{pmatrix} 9 & 4 \\ 4 & 9 \end{pmatrix}$$

and obtain its orthogonal diagonalization $V\Lambda V^T$. We find the eigenvalues solving
$\det(A^TA-\lambda I)=0\Rightarrow(9-\lambda)^2-16=0\Rightarrow\lambda=5,13$. The normalized eigenvectors are found for $\lambda=5$:

$$(A-5I)\underline{\mathbf{v}}=\underline{\mathbf{0}}\Rightarrow\begin{pmatrix}4&4\\4&4\end{pmatrix}\underline{\mathbf{v}}=\underline{\mathbf{0}}\Rightarrow\underline{\mathbf{v}}_1=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

and for $\lambda=13$:

$$(A-I)\underline{\mathbf{v}}=\underline{\mathbf{0}}\Rightarrow\begin{pmatrix}-4&4\\4&-4\end{pmatrix}\underline{\mathbf{v}}=\underline{\mathbf{0}}\Rightarrow\underline{\mathbf{v}}_2=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$$

So we have

$$V=\frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\-1&1\end{pmatrix}\Rightarrow A^TA=\begin{pmatrix}9&4\\4&9\end{pmatrix}=\frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\-1&1\end{pmatrix}\begin{pmatrix}5&0\\0&13\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1&-1\\1&1\end{pmatrix}=V\Lambda V^T,$$

the orthogonal diagonalization of A^TA. The singular values are the roots of the eigenvalues so

$$\Sigma=\begin{pmatrix}\sqrt{5}&0\\0&\sqrt{13}\\0&0\end{pmatrix}$$

To find $\underline{\mathbf{u}}_i$ we use $\underline{\mathbf{u}}_i=\frac{1}{\sigma_i}A\underline{\mathbf{v}}_i$ and get

$$\underline{\mathbf{u}}_1=\frac{1}{\sqrt{5}}\begin{pmatrix}1&2\\2&-1\\2&2\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}=\frac{1}{\sqrt{10}}\begin{pmatrix}-1\\3\\0\end{pmatrix}\text{ and }\underline{\mathbf{u}}_2=\frac{1}{\sqrt{13}}\begin{pmatrix}1&2\\2&-1\\2&2\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}=\frac{1}{\sqrt{26}}\begin{pmatrix}3\\1\\4\end{pmatrix}.$$

Finally, to find the third column of U, we solve

$$A^T\underline{\mathbf{u}}_3=\underline{\mathbf{0}}\Rightarrow\begin{pmatrix}1&2&2\\2&-1&2\end{pmatrix}\underline{\mathbf{x}}=\underline{\mathbf{0}}.$$

A single row operation gives $5y=-2z$ and $x+2y=-2z$, so after normalization, we have

$$\underline{\mathbf{u}}_3=\frac{1}{\sqrt{65}}\begin{pmatrix}6\\2\\-5\end{pmatrix}\Rightarrow U=\begin{pmatrix}-\frac{1}{\sqrt{10}}&\frac{3}{\sqrt{26}}&\frac{6}{\sqrt{65}}\\\frac{3}{\sqrt{10}}&\frac{1}{\sqrt{26}}&\frac{2}{\sqrt{65}}\\0&\frac{4}{\sqrt{26}}&-\frac{5}{\sqrt{65}}\end{pmatrix}$$

The SVD is therefore

$$A=\begin{pmatrix}1&-1&0\\-1&0&1\end{pmatrix}=\frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\-1&1\end{pmatrix}\begin{pmatrix}\sqrt{3}&0&0\\0&1&0\end{pmatrix}\begin{pmatrix}\frac{2}{\sqrt{6}}&-\frac{1}{\sqrt{6}}&-\frac{1}{\sqrt{6}}\\0&-\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{3}}\end{pmatrix}=U\Sigma V^T$$